

Interdisciplinary

The Limits and Challenges of Particle Physics: Critiquing Contemporary Research Methods

Nicolas A. R. Mousset

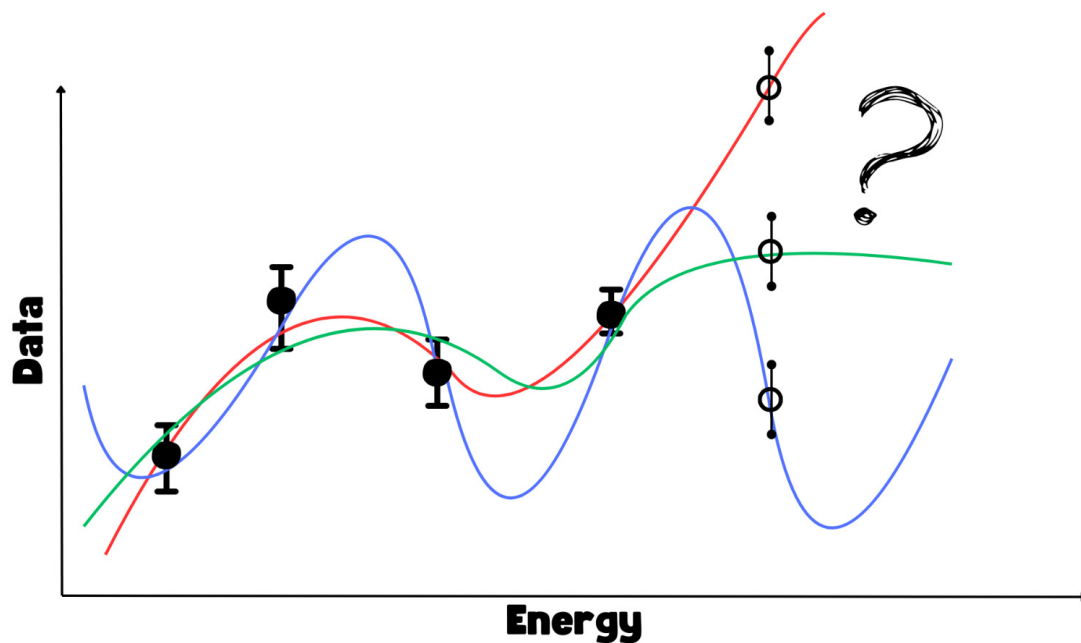


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Suggested citation:

Mousset, N. A. R. (2024). The Limits and Challenges of Particle Physics: Critiquing Contemporary Research Methods. *AUC Undergraduate Journal of Liberal Arts & Sciences*, 22(1), 32–41.

1 Introduction

The development of particle physics models has been very successful both in its predictive nature and in applications to society. It aided humanity in the battle against cancer, allowed scanning cargo for contraband, stream-lined processes in computer science, and led to the development of new kinds of epoxies and plastics, just to name a few applications (United States/Large Hadron Collider, n.d.). However, this particular branch of physics hasn't seen any major developments since the popularisation and adoption of the Standard Model of Particle Physics in the 1970s: the most successful standardised theory of physics describing interactions between particles. Since then, particle physicists were only able to confirm the predictions of the Standard Model. While this certainly indicates that the theory is successful in current experimental set-ups, it also demonstrates a lack of progress in the field. Without the ability to develop and improve scientific models, the applications and experimental value of scientific models will no longer improve, which has undesirable consequences discussed further. Additionally, I am of the opinion that it is important to address such accusations seriously and, if they are true, try to resolve the issues.

In this essay, I analyse the criticism put forward by Dr. Sabine Hossenfelder in her YouTube video "What's Going Wrong in Particle Physics? (This is why I lost faith in science.*)" released on the 11th of February 2023, which attracted more than 1,5 million views in the year after publication. Dr. Hossenfelder has a PhD in physics and has contributed to more than 80 research papers on particle physics, astrophysics, quantum foundations, and more ("Research," 2022). She is thus qualified to voice criticisms in her field. Since, she has become a popular science communicator on Youtube, where she discusses scientific papers and issues dear to her heart.

Dr. Hossenfelder criticises the scientific method of certain particle physicists and argues that they are slowing down progress and not performing science. Further, she asserts that physicists exploit the principles of Underdeterminism and Incommensurability: Underdeterminism describes the epistemological fact that a limited amount of knowledge supports multiple theories, and Incommensurability is the issue of

non-objective standards to compare the theories. These are usually limitations on the development of scientific knowledge, yet some physicists may use them to their advantage. I will investigate her criticisms with the help and insight of another physicist, Prof. Dr. Thomas Schmidt.

In order to research for this paper, I contacted Prof. Schmidt, who is working as a professor of physics at the University of Luxembourg and has a PhD in physics ("Prof. Dr. Thomas Schmidt," n.d.). I corresponded with him in autumn of 2023, and he gave me in-depth insight into the thinking of a working physicist. He will provide the position of an experienced researcher to allow me to properly evaluate Dr. Hossenfelder's criticisms towards the field they are both involved in.

I will begin by giving a brief overview of the history of particle physics, followed by a discussion of the criticisms put forth by Dr. Hossenfelder in her video, including reflections from both Prof. Schmidt and myself. After analysing the four main criticisms and capturing the essence of Dr. Hossenfelder's sentiment, I will reflect on the arguments regarding the current research methods, and provide a perspective for the future.

2 A Brief History of Particle Physics

Physicists have developed categories for the forces of nature. They include gravity, the strong nuclear force, the weak nuclear force, and the electromagnetic force. The general goal is to understand the forces better and, if possible, reduce the number of distinct natural forces or even combine them all into a single force. Gravity was very successfully described by Albert Einstein's theories of general (1905) and special relativity (Einstein, 1915/1997). In the 1950s, Yang and Mills (1954) developed a model to accurately describe strong nuclear interaction. In the 1960s, this model was extended by Sheldon Glashow (1961) reducing the electromagnetic force and the weak nuclear force to a single force and model. Finally, when Weinberg and Salam (independently) introduced the Higgs boson (one of the elementary particles fundamental for the development of Particle Physics) to the model of Glashow's model of electroweak interaction, it was prestigiously named the Standard Model of Particle Physics and

remained unchanged since that paper in 1967 (“Weinberg, Salam and Glashow,” 1980).

Nowadays, the experimental research that is being done often confirms the Standard Model and as of now no clear anomalies have been identified that could indicate that the theory is not accurate enough. Some particle physicists’ research consists in solving specific problems of the Standard Model, which are not considered important by other particle physicists, specifically Dr. Hossenfelder. As I am focusing on Dr. Hossenfelder’s criticism of this research, in the following sections, I will focus on some of the examples she uses in her video.

One of these examples used by Dr. Hossenfelder is the idea of supersymmetry (SUSY). In 1966, the idea of SUSY was proposed as a hypothetical symmetric mechanism of particle physics (Dimopoulos & Georgi, 1981). It was introduced in order to solve the so-called Hierarchy Problem, which questions why the mass of the Higgs boson is so small (Strassler, 2011). This is seen as a problem by some particle scientists because the mass of the Higgs boson seems to be balanced incredibly precisely considering the enormous quantum corrections involved. The values involved are as delicate as balancing a pencil on its tip, if the pencil was the length of the entire solar system (Giudice, 2008, pp. 162-163). Such an extreme precision is claimed to be “unnatural” and thereby worthy of investigation in the opinion of some physicists. Additionally, physicists raise the question to why the mass of the Higgs boson is so small (125 GeV) compared to the Planck scale (10^{19} GeV) and so close to the weak-nuclear force (100 GeV). This is puzzling to some physicists who would prefer to have a mechanism in place that explains the stable energy state of the Higgs boson. As of right now, the mass of the Higgs boson is not controlled or protected by any symmetry. SUSY is a contender for a symmetry that would protect the Higgs boson mass from changing and forcing it to remain stable and small. Some physicists also solved this “problem” by introducing more dimensions to the standard model of particle physics.

Another proposed theoretical particle is the Axion, introduced in 1977 and named after a brand of laundry detergent for how “clean” of a solution it was to the Strong CP Problem. This problem questions why the theta parameter (a con-

stant of the Standard Model that must be experimentally determined) is so small (Chadha-Day et al., 2022). This situation is very similar to the Hierarchy problem as both ask why a parameter is so small. Theta would be expected to be around 1 by conventions but it’s close to 0 (yet there is no physical mechanism telling us it should be larger than this). Still, some physicists can’t stand exceedingly small numbers without a system controlling the values, keeping them small.

Finally, I want to introduce the idea of Proton Decay, which is exactly what it sounds like. It proposes that protons exhibit particle decay, which allows for breaking baryon number symmetry (a type of conservation principle of the Standard Model) and extends the Standard Model in such a way that it supports certain Grand Unified Theories (Ohlsson, 2023). Grand Unified Theories are inspired by an aesthetic ideal, they describe theories that combine the electroweak interaction and the strong nuclear force into a singular force. Reducing the number of distinct forces would allow physicists to explain even more phenomena from the same basic mechanisms.

3 Addressing Dr. Hossenfelder’s Criticism

3.1 Pseudo Problems and Real Problems

The first criticism Dr. Hossenfelder puts forward in her video is that particle physicists are currently focusing on the wrong problems, which they try to solve by introducing new theoretical particles. These theories however are extensions of the Standard Model in such a way that they agree with all existing experimental data but also predict a certain observation in an untested energy range.

Dr. Hossenfelder distinguishes between “Pseudo Problems” and, for the purposes of this paper, “Real Problems.” She did not outright call them “Real Problems” but I need a contrasting term. She defines “Real Problems” clearly: either the current model is inconsistent with new experimental data which is peer-reviewed and significant, or there are internal inconsistencies discovered which allow for contradictory predictions of the model. If neither of these two

criteria apply, then the issue falls under “Pseudo Problems” which are thus defined as the negative of “Real Problems” (Hossenfelder, 2023).

Dr. Hossenfelder accuses current research of focusing on these “Pseudo Problems” which, according to she, include the aforementioned examples. Grand Unified Theories are not a necessary change to the Standard Model as they do not solve “Real Problems.” Why the mass of the theta parameter of the Standard Model or the mass of the Higgs boson are so minuscule is also not a “Real Problem” following Dr. Hossenfelder’s definition. Similar to the question why the speed of light is exactly c , it is not motivated by an internal inconsistency or inconsistent predictions (contradictions of the model) and therefore not a “Real Problem.” Only if our research confirmed that the speed of light as we think of it creates paradoxes in our theories or contradictions with observations would it become a “Real Problem.”

Looking at the specific examples introduced earlier, the Hierarchy problem is not a “Real Problem” because it is not motivated by either an inconsistency in the model nor a contradiction with data. It is simply a case of scientists asking why a parameter is small, and how it remains small. To solve this “issue” they introduce more particles that are never observed, new dimensions specifically just to tackle this issue, or new mechanisms and symmetries that have no experimental evidence to date or are borderline not falsifiable. Proton decay, for instance, which is motivated by the idea of reducing the amount of equations from the standard model in favour of GUTs, currently estimates the half-life of a proton to be around 10^{32} years (Nave, n.d.).

Dr. Hossenfelder argues particle physicists are guided towards “Pseudo Problems” by false ideals such as aesthetics, elegance, or intuition. They wonder why the Standard Model does not explain why a number is what it is, or how they can extend the Standard Model to conform to any specific kind of symmetry they prioritise, but their efforts are seemingly non-scientific and led by personal preference.

There were extensive studies done about the extensions of the Standard Model. These models predicted certain particles to exhibit a specific behaviour at high energy levels that could be measured. However, no such predicted particles have been observed so far. To give a few examples:

light fermionic dark matter (Fadelli, 2022), sterile neutrinos (Feuerstein, 2023), and Light WIMPs (Yang et al., 2019). No effects predicted by the respective models have been observed.

The proposal of these particles can therefore be criticised, but Dr. Hossenfelder offers a good counter argument for possible objections — “While they are working on it (any given Pseudo Problem) they might stumble over something else that’s interesting.” (2023, 17:09). However, she still argues such research would constitute a waste of funds and immensely cripple the speed of progress. As a solution, Dr. Hossenfelder suggests that particle physicists focus on “Real Problems” by identifying them using the aforementioned criteria, and abandoning “Pseudo Problems.”

I am convinced that Dr. Hossenfelder’s criticism stands, and indeed, so is Prof. Schmidt: “*Extending it [The Standard Model of particle Physics] has become a ‘free for all’, where theorists can propose any number of crazy extensions, because the energies required to falsify these predictions would be inaccessibly high*” (T. Schmidt, personal communication, November, 2023).

He adds that,

I generally agree with her propositions as far as they concern particle physics. Luckily in most other domains, there is a much closer relationship between theory and experiment, such that theories can be falsified more easily. Therefore, reactions like “I lost my faith in science” are nonsense, unless we equate “science” with “particle physics”, which Hossenfelder seems a bit prone to do. (T. Schmidt, personal communication, November, 2023)

Particle physics has reached a point where it can no longer apply established methods to promote new discoveries. In Prof. Schmidt’s experience, this is close to unique for particle physics and most scientific areas of study do not have this issue. It seems that the focus on “Pseudo Problems” is indeed a problem for the field, not “science” in general.

3.2 The Role of Aesthetics

One of the main guiding factors for particle physics is aesthetics, or the idea of beauty. The idea that a better theory will be more “elegant” or “beautiful” in the sense that it will be very abstract, self-contained, and self-explanatory. To quote Prof. Schmidt:

The problem with the standard model seems to be that it is not "beautiful." It's true that it rests on actually very few equations but many parameters enter this theory which can only be determined experimentally. Einstein's theory, on the other hand, only depends on the speed of light and the gravitational constant (which makes it fit the ideal of aesthetics in physics). (T. Schmidt, personal communication, November, 2023)

Dr. Hossenfelder argues that beauty is an unjustified and arbitrary ideal and that it is not a valid tool for the search of new models as they will be manifestations of human biases that lead physicists to work on “Pseudo Problems.” Prof. Schmidt agrees again with Dr. Hossenfelder on this point:

most are guided by the principle of "beauty", i.e., that a new theory must somehow be elegant. This has worked in the past, but of course there is a priori no reason to believe that new theories must be beautiful. Logically speaking, she is right about this point, but she offers no alternative strategy for developing new theories. (T. Schmidt, personal communication, November, 2023)

I would argue that Dr. Hossenfelder suggests identifying “Real Problems” and abandoning “Pseudo Problems” as an alternative strategy for developing new theories, but I would have to concede that she does not offer a concrete method on how to conduct science without somewhat arbitrary ideals such as aesthetics. It is possible that the discussion of such a complex issue did not fit into a 20 minute YouTube video.

Critiquing the ideal of aesthetics is justified in my opinion, and I agree with the statement

made by Prof. Schmidt. In my own opinion, aesthetics should never be the motivation for research. Einstein’s theory of relativity was particularly “beautiful” to the point that Einstein himself was surprised, but there is no reason to believe that theories should be beautiful or should have a low amount of experimentally determined values. I would suggest focusing on “Real Problems” and the related criteria to determine on which research should be prioritised. The role of aesthetics would be reduced but it could still play a role when deciding between two models of equivalent predictabilities. For instance, much like Occam’s Razor, I would prefer that aesthetics may only be used when deciding between equivalent theories, which should not discredit the other theory, but may cause it to be less popular. This would reduce the impact of arbitrary ideals on our research question while keeping the ideal of elegance and beauty when it comes to practical applications. Thus, I believe it could be possible to conduct research without the ideal of aesthetics, while focusing on the inconsistencies in the data and models instead.

3.3 The Social and Financial Impact

I already touched upon the criticism that, according to Dr. Hossenfelder, research dedicated to “Pseudo Problems” is a waste of funds. Not only could the researchers working on them be working on “Real Problems” instead, but the experiments they are conducting are enormously expensive, outweighing the benefit of knowledge that can be gained from them, if there is any (Researchers Awarded \$3.7M to Build Next-Generation Halo-scope in Search for Dark Matter, n.d.). Additionally, many proposed models require the construction of larger experimental research facilities, including the Future Circular Collider (FCC) which will be 90 km long and will supposedly cost 22 billion USD (Castelvecchi & Gibney, 2020). The most powerful particle collider to date and currently in use is the Large Hadron Collider (LHC) which is 27 km long, and cost about 4,75 billion USD to build.

Prof. Schmidt describes the issue well:

I don't think that the current experiments based on particle accelerators are worth pursuing until we know better where to look. People haven't

found any signs of supersymmetry (one cornerstone, e.g., of string theory) at LHC. With tremendous effort one might be able to reach 10 times the energy of LHC, but it's not clear that new physics should emerge at this scale and not at energies, say, a billion times higher, which will remain inaccessible using these machines. Hence, I think that in this particular field, the impetus has to come from somewhere else. (T. Schmidt, personal communication, November, 2023)

Dr. Hossenfelder also mentions that a considerable risk of pursuing “Pseudo Problems” lies in inefficient budget use of mostly tax payer money within the field of particle physics. When most particle physics researchers are focusing on “Pseudo Problems” and cannot produce significant results, yet demand more financial support to conduct more expensive experiments focusing on the same problems, a lot of money invested in particle physics research will be wasted. The inefficient use of budget within the field of particle physics could easily lead to a lack of interest by the academic community; when a research field has not developed their theory and not turned up any major results in 40 years, universities and governments will begin redistributing their research investments. The field of particle physics will have less and less ability to gain research grants or find institutes willing to invest large sums of money in a field that has seemingly few promises: a view that has been skewed by the “Pseudo Problems” not being worth the financial investment.

A perfect example of this is the FCC. While it is estimated to cost 22 billion USD, the researchers advocating for its construction do not actually have any data justifying the FCC. They are essentially saying: “We haven’t found any evidence for our theories but perhaps if we spend another few billions on new laboratory equipment and research facilities we can test our theory.” This is not a very convincing research proposal.

Additionally to the risk of a disinterested academic community, particle physics could face a lack of interest of the general public and most importantly students considering a career in particle physics. If this should happen, it will create a feedback loop between particle physics becom-

ing less productive and less people deciding to become particle physicists. The end of the feedback loop could be the end of particle physics research according to Dr. Hossenfelder.

I would argue this feedback loop could only take form if the researchers never realised that their pursuit of “Pseudo Problems” is not going anywhere. I am optimistic that it is only a question of when the researchers will realise this. If they do, it is to hope that they focus the efforts of particle physics on developing new methods and modes of thinking. Therefore, I think Dr. Hossenfelder’s concerns are to some degree exaggerated. However, given that the decrease in funding due to slow progress could be a serious issue, her prediction could also be more accurate than it seems.

We can observe some circumstances like this already. As mentioned earlier, SUSY was first proposed in 1966 and since then no experimental evidence was produced in favour of the theory. As of right now, SUSY is merely an extended model of particle physics (one of many) and aligns with the Standard Model of particle physics in all observations, but also makes predictions about measurements we are not able to do.

This comes back to the epistemological issue of Underdeterminism—when we don’t have enough data to decide between theories because for our limited data all models behave the same—and Incommensurability—we do not have any objective standard to compare theories that behave the same way. It is unclear whether the researchers are making use of these restrictions on knowledge on purpose to generate invalid research proposals or grants. Either way, Dr. Hossenfelder argues their practice is non-scientific, as will be discussed in the next section.

3.4 Scientific Validity and Paradigm Shifts

The final criticism Dr. Hossenfelder presents is the accusation that the current research trend in particle physics is not “science” according to modern criteria. Dr. Hossenfelder states that a good way of seeing that modern particle physics research is not scientific, is that “it is not working.” However, though it is true that there is no progress in the Kuhnian sense of paradigm shifts, this also implies that the Standard Model is an im-

mensely successful theory.

It becomes clear their science is not working when considering Karl Popper's philosophy of science, in which he states, among other principles, that a scientific theory must be abandoned after it has been falsified (Shea, n.d.). Arguably the existence of particles cannot be falsified, yet the presence of certain phenomena is predicted by the existence of these particles, and these phenomena are not observed experimentally. This leads to the physicists "abandoning" their theory and postulating a "new" one which is very similar to the "old." This new theory postulates the exact same particles and phenomena, only at higher energies that are inaccessible to current experimental set-ups. This means the experiments require the construction of larger research facilities to reach the conditions they claim to be required to falsify the theory. One could argue that these physicists are not abandoning their theories but adapting them. This process was heavily criticised by Popper employing his criteria of non-science.

Similarly, other models that evade falsification by adaptation are likely unscientific, such as Astrology, which is able to explain any event in many different ways. A person engaging in astrology will be able to explain any behaviour a person exhibits through their star-sign, even contradictory statements. A theory that can explain everything cannot predict anything as everything can be explained with the theory. The fact that astrology is therefore unable to predict people's behaviour through the knowledge of their star sign alone, and that astrologers can easily change their claims using post hoc changes to their theory by picking and choosing which aspects they take into account implies that astrology cannot be classified as a science. Therefore, I can agree that these particular models and how they are adapted to continue existing is non-scientific. Yet, it is clear that this is not the case for all of particle physics research. Some theories are changed more dramatically or even abandoned, of course, never by all particle physicists. For example, the Axion particle was introduced in the 70s as the "Harmless Axion," then experimentally ruled out, and rebranded as the "Invisible Axion." Today, it is just known as the "Axion."

A model that makes perfect predictions but is incapable of predicting other phenomena, such as the Standard Model, is paradoxically problematic

because of its success. As Prof. Schmidt puts it: *"We have in fact two immensely successful theories, the standard model and the general theory of relativity, but unfortunately they don't connect. A theory which agrees with all experiments is really the worst case if you want to make progress in fundamental research"* (T. Schmidt, personal communication, November, 2023).

Currently, particle physics is in the "normal science" stage of the Kuhnian paradigm cycle, where scientists work with an established paradigm, in this case the Standard Model, and apply it to problems and research set-ups. The next stage in the paradigm cycle is the "model shift" stage, which would be triggered by anomalies found during experiments (Bird, 2022). Dr. Hossenfelder would constitute this as a "Real Problem" since this would constitute inconsistency with new data. Such a development could allow particle physics to enter the next Kuhnian paradigm stage, the "model drift" stage, and equate to progress in the field.

As of right now, the Standard Model is so successful that no anomalies have been clearly identified which could be grounds for more rigorous and specific research. Prof. Schmidt agrees with this sentiment: *"The greatest discovery would be an experimental result which contradicts the standard model. There are some hints of this, e.g., in ' $g - 2$ ' and some other experiments, but it could still be a statistical error"* (T. Schmidt, personal communication, November, 2023). The Muon $g - 2$ ("gee minus two") experiment is performed at the Fermi National Accelerator Laboratory, or Fermilab, and intends to measure a specific numerical parameter in particle physics: g (not to be confused with the gravitational constant). G is only a little larger than 2, so the scientific community evaluates the value of $g - 2$ to work with one very small number and determine its deviation from 0. This small number of $g - 2$ is predicted by the Standard Model and, as of right now, is a candidate for being an anomaly (Castelvecchi, 2023). However, further studies which measure $g - 2$ with greater precision and lower error margins are required before it presents a "Real Problem" for the Standard Model.

I would argue that particle physics, to become more productive, needs an anomaly to trigger a paradigm shift and introduce new priorities that would out-rank aesthetic ones. Although aesthet-

ics would still play a role in determining what research is seen as valuable, it would allow for a clearer view of “Real Problems” in the field. Prof. Schmidt

fully agree[s], and [...] would say we need a new paradigm. The standard model is an ‘effective field theory’ which is based on certain symmetries of the fundamental forces we observe at the currently accessible energies. New theories arise from extrapolating this to higher energies. It is generally assumed that the $U(1) \times SU(2) \times SU(3)$ symmetry of the standard model arises as a broken symmetry state of a more symmetric theory at higher energies. But there are just too many options for such a theory. People talk about ‘string theory’, and it might well be the solution, but it is just a term for a vast number of possible theories. But there are certain aspects that all such theories should share, e.g., extra dimensions, new particles, so the search is not hopeless. It is just extremely hard with current experimental setups. (T. Schmidt, personal communication, November, 2023)

In December of 2023, a team of researchers cooperating globally in many research facilities published a summarising paper including a precise measurement of $g - 2$ (Benton et al., 2023). They kindly included an informative figure:

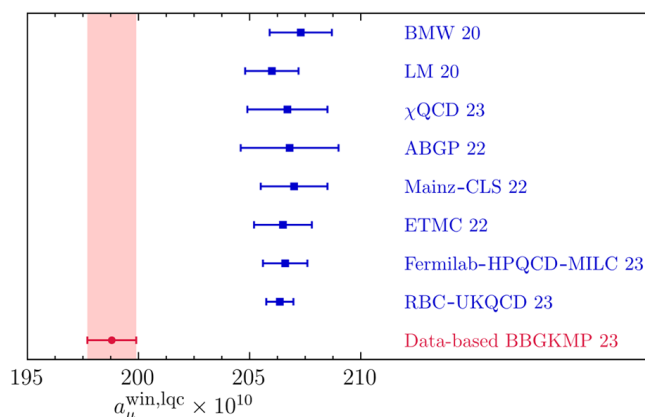


Fig. 1 taken from Benton et al., 2023.

This figure shows in blue different theoretical value predictions made by different extensions

and variations of the standard model of particle physics. The measured data is featured in red, including error bars for all theories and measurements. The certainty of this study is self-reported to be between 3.3σ and 6.1σ , or a chance of roughly 3.6% to 0.0003% that the results are coincidental (Benton et al., 2023). The golden standard of certainty for physics is 6 sigma, leaving only a 0.0004% chance that the results are caused by a random variation. In other words, this study has a really high chance of clearly laying out an anomaly in the Standard Model, allowing for more specialised research focusing on this anomaly.

4 Conclusion and Future Prospects

In conclusion, Dr. Hossenfelder’s criticism towards current research in particle physics is, in my opinion, valid to an extent. However, as she is quite passionate about the topic, she dramatises and exaggerates a little at some points. Her definitions of “Pseudo Problems” and “Real Problems” are clear and instrumental when it comes to the categorisation of problems in particle physics. Her critique of aesthetics is valid and justified, yet she misses out on providing an alternative. I do not see this as a major objection since philosophical reasoning can be applied to a problem and Dr. Hossenfelder’s definitions to determine its nature. Additionally, her criticism that these experiments looking for new particles and the testing of the models is budget-inefficient and could lead to particle physics’ progress slowing down further, or even dying out, applies in both my and Prof. Schmidt’s opinion. Her accusation that most current research in particle physics is not “science” is controversial but, in the end, I would agree with her statement though not in its totality as there is research being done on suspected “Real Problems,” such as $g-2$. Although, I would agree that there is a lot of research being done that follows the pattern outlined in paragraph 2 of section 3.4) which is non-scientific. Therefore, I am of the opinion that her criticisms apply to particle physics to some extent, but that the situation is not as dire as she presents it to be.

Recently, Aniculaesei et al. (2023) published a paper on a new kind of particle accelerator that produces 100 times more energy per length

than the LHC, which could prove useful in speeding up the process of finding anomalies. Even though this exploration of philosophy regarding particle physics made me more sceptical of physical models, I remain hopeful that humanity will overcome the inefficient research methods employed by some researchers. Even if progress slows down, it will eventually continue. I still see the success of the Standard Model not just as a symbol of stagnation, but also as one of human triumph in conceptual construction. We are already able to use the model to accurately describe and predict most phenomena that we have observed. This is a true achievement, seeing as they are but human ideas developed on the basis of other human ideas and observations. We will reach the limit of our abilities regarding the development of scientific theories as we approach the limits of observation. Some believe that we have reached that limit. Some such as John Horgan who, in his book "The End of Science: Facing the Limits of Knowledge," explains how humans necessarily have a limit to their ability to abstract and develop more and more complex theories. It is possible that in the future, a consistent and perfect Grand Unified Theory will be proposed by ChatGpt20. Then the question becomes: Will we be able to understand it?

References

- Aniculaesei, C., Ha, T., Yoffe, S., Labun, L., Milton, S., McCary, E., Spinks, M. M., Quevedo, H. J., Labun, O. Z., Sain, R., Hannasch, A., Zgadzaj, R., Pagano, I., Franco-Altamirano, J. A., Ringuette, M. L., Gaul, E., Luedtke, S. V., Tiwari, G., Ersfeld, B., ... Hegelich, B. M. (2024). The acceleration of a high-charge electron bunch to 10 GeV in a 10-cm nanoparticle-assisted wakefield accelerator. *Matter and Radiation at Extremes*, 9(1). <https://doi.org/10.1063/5.0161687>
- Benton, G., Diogo Boito, Golterman. M., Keshavarzi, A., Maltman, K., & Peris, S. (2023). Data-Driven Determination of the Light-Quark Connected Component of the Intermediate-Window Contribution to the Muon $g - 2$. *Physical Review Letters*, 131(25), 251803–251803. <https://doi.org/10.1103/physrevlett.131.251803>
- Bird, A. (2022). *Thomas Kuhn*. The Stanford Encyclopedia of Philosophy. <https://plato.stanford.edu/entries/thomas-kuhn/>
- Castelvecchi, D. (2023). Dreams of new physics fade with latest muon magnetism result. *Nature (London)*, 620(7974), 473–474. <https://doi.org/10.1038/d41586-023-02532-6>
- Castelvecchi, D., & Gibney, E. (2020). CERN makes bold push to build €21-billion supercollider. *Nature (London)*. <https://doi.org/10.1038/d41586-020-01866-9>
- "Weinberg, Salam and Glashow on physics. (1980)." *CERN Courier*, 20(8), 350-357.
- Chadha-Day, F., Ellis, J., & Marsh, D. J. E. (2022). Axion dark matter: What is it and why now? *Science Advances*, 8(8). <https://doi.org/10.1126/sciadv.abj3618>
- Dimopoulos, S., & Georgi, H. (1981). Softly broken supersymmetry and SU(5). *Nuclear Physics B*, 193(1), 150–162. [https://doi.org/10.1016/0550-3213\(81\)90522-8](https://doi.org/10.1016/0550-3213(81)90522-8)
- Einstein, A. (1905). Zur Elektrodynamik bewegter Körper. *Annalen der Physik*, 322(10), 891–921. <https://doi.org/10.1002/andp.19053221004>
- Einstein, A. (1997). The Field of Equations of Gravitation. In A. Engel (Trans.), *The Collected Papers of Albert Einstein* (Vol. 6,

- pp. 117–120). Princeton University Press.
- Fadelli, I. (2022, November 14). *Recent searches for light fermionic dark matter by the PandaX-4T collaboration*. Phys.org. <https://phys.org/news/2022-11-fermionic-dark-pandax-4t-collaboration.html>
- Feuerstein, B. (2023, January 11). *Final results from the STEREO experiment reject sterile neutrino hypothesis*. Phys.org. <https://phys.org/news/2023-01-results-sterео-sterile-neutrino-hypothesis.html>
- Giudice, G. F. (2008). Naturally Speaking: The Naturalness Criterion and Physics at the LHC. In G. Kane & A. Pierce (Eds.), *Perspectives on LHC Physics* (pp. 155–178). World Scientific Publishing Company. https://doi.org/10.1142/9789812779762_0010
- Glashow, S. L. (1961). Partial-symmetries of weak interactions. *Nuclear Physics*, 22(4), 579–588. [https://doi.org/10.1016/0029-5582\(61\)90469-2](https://doi.org/10.1016/0029-5582(61)90469-2)
- Hossenfelder, S. [Sabine Hossenfelder]. (2023, February 11). *What's Going Wrong in Particle Physics? (This is why I lost faith in science.)* [Video]. YouTube. <https://youtu.be/lu4mH3Hmw2o>
- Nave, R. (n.d.). *Protons and neutrons*. HyperPhysics. <http://hyperphysics.phy-astr.gsu.edu/hbase/Particles/proton.html>
- Ohlsson, T. (2023). Proton decay. *Nuclear Physics B*, 993, 116268. <https://doi.org/10.1016/j.nuclphysb.2023.1162680>
- Prof. Dr. Thomas Schmidt. (n.d.). Université du Luxembourg. <https://www.uni.lu/en/person/NTAwMDg00TdfX1Rob21hcyBTQ0hNSURU/>
- Researchers awarded \$3.7M to build next-generation haloscope in search for dark matter. (n.d.). Berkeley Engineering. <https://engineering.berkeley.edu/news/2023/12/researchers-awarded-3-7m-to-build-next-generation-haloscope-in-search-for-dark-matter/>
- Research. (2022, November 21). Sabine Hossenfelder. <https://sabinehossenfelder.com/research-2/>
- Shea, B. (n.d.). *Popper, Karl: Philosophy of Science*. Internet Encyclopedia of Philosophy. <https://iep.utm.edu/pop-sci/#H2>
- Strassler, M. (2011, August 16). The Hierarchy Problem. *Of Particular Significance*. <https://profmattstrassler.com/articles-and-posts/particle-physics-basics/the-hierarchy-problem/>
- Yang, C. N., & Mills, R. L. (1954). Conservation of Isotopic Spin and Isotopic Gauge Invariance. *Physical Review*, 96(1), 191–195. <https://doi.org/10.1103/physrev.96.191>
- Yang, L. T., Li, H. B., Yue, Q., Ma, H., Kang, K. J., Li, Y. J., Wong, H. T., Agartioglu, M., An, H. P., Chang, J. P., Chen, J. H., Chen, Y. H., Cheng, J. P., Deng, Z., Du, Q., Gong, H., Guo, Q. J., He, L., Hu, J. W., ... Zhu, Z. H. (2019). Search for Light Weakly-Interacting-Massive-Particle Dark Matter by Annual Modulation Analysis with a Point-Contact Germanium Detector at the China Jinping Underground Laboratory. *Physical Review Letters*, 123(22), 221301–221301. <https://doi.org/10.1103/PhysRevLett.123.221301>
- United States/Large Hadron Collider. (n.d.). *In Benefits of Particle Physics* [Fact Sheet]. U.S. Department of Energy & National Science Foundation. https://www.fnal.gov/gri-dfest/pdfs/benefits_factsheet.pdf